Galatea: Supplementary Material

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Equation, proposition and section numbers refer to the main paper.

Appendix: Proofs

Proof of Proposition 1: The objective of (4) is strictly concave (the entropy is strictly concave and the KL term is concave in $\mathbf{P}$ because it is the negative of a convex function of the linear map $\mathbf{P} \mapsto \mathbf{P}^\top \mathbf{1}$). The feasible set is convex and non-empty, and the objective is coercive because $-P \log P$ dominates the linear term, so the maximiser exists and is unique. The entropy forces $P_{ij} > 0$, so only the equality constraints are active. With multipliers $-f_i/\varepsilon$ for the rows and $\mathbf{q} = \mathbf{P}^\top \mathbf{1}$, stationarity gives $\bar{S}_{ij} - \varepsilon \log P_{ij} - \tau \log(q_j/b_j) + f_i = 0$, that is, $P_{ij} = \exp\big((\bar{S}_{ij} + f_i + g_j)/\varepsilon\big)$ with $g_j := \tau \log(b_j/q_j)$. Summing over i gives $q_j = e^{g_j/\varepsilon} \sum_i e^{(\bar{S}_{ij}+f_i)/\varepsilon}$. Substituting into the definition of g_j gives $g_j(1 + \tau/\varepsilon) = \tau[\log b_j - \mathrm{LSE}_i((\bar{S}_{ij} + f_i)/\varepsilon)]$, which is (6) with $\lambda = \tau/(\tau + \varepsilon)$. The row constraint $\sum_j P_{ij} = a_i$ is (5). (i) Let Φ be the composed map of Proposition 2, which is λ-Lipschitz. At the fixed point, $\|\mathbf{g}^\star\|_\infty \le \|\Phi(\mathbf{g}^\star) - \Phi(\mathbf{0})\|_\infty + \|\Phi(\mathbf{0})\|_\infty \le \lambda \|\mathbf{g}^\star\|_\infty + \|\Phi(\mathbf{0})\|_\infty$, and $\|\Phi(\mathbf{0})\|_\infty = O(\lambda)$ by (6). Hence $\|\mathbf{g}^\star\|_\infty \le \|\Phi(\mathbf{0})\|_\infty/(1-\lambda) = O(\lambda)$, so $\mathbf{g}^\star \to \mathbf{0}$ as $\lambda \to 0$. Substituting $\mathbf{g} = \mathbf{0}$ into (5) gives $P_{ij} = a_i e^{\bar{S}_{ij}/\varepsilon} / \sum_k e^{\bar{S}_{ik}/\varepsilon}$. At $\tau = 0$ this holds exactly, because the problem decouples over rows into independent entropy-regularised linear programs on scaled simplices. (ii) As $\tau \to \infty$, $\tau\,\mathrm{KL}(\cdot\|\mathbf{b})$ Γ-converges to the indicator of $\{\mathbf{P}^\top \mathbf{1} = \mathbf{b}\}$. Since the objectives are equi-coercive and the maximisers unique, $\mathbf{P}^\tau \to \mathbf{P}^{\mathrm{bal}}$ [1]. Setting $\lambda = 1$ in (5)–(6) gives the balanced Sinkhorn equations. □

Proof of Proposition 2: For any vector $\mathbf{x}$ and perturbation $\boldsymbol{\delta}$, $|\mathrm{LSE}(\mathbf{x} + \boldsymbol{\delta}) - \mathrm{LSE}(\mathbf{x})| \le \|\boldsymbol{\delta}\|_\infty$, because $\nabla \mathrm{LSE}$ is a probability vector. Hence the map $\mathbf{g} \mapsto \mathbf{f}(\mathbf{g})$ of (5) is 1-Lipschitz in $\|\cdot\|_\infty$, and the map $\mathbf{f} \mapsto \mathbf{g}$ of (6) is λ-Lipschitz. Their composition Φ is λ-Lipschitz. For $\lambda < 1$ it is a contraction on $(\mathbb{R}^{m+1}, \|\cdot\|_\infty)$, so by Banach's theorem it has a unique fixed point $\mathbf{g}^\star$ (the potential of Proposition 1), and $\|\mathbf{g}^{(L)} - \mathbf{g}^\star\|_\infty \le \lambda^L \|\mathbf{g}^{(0)} - \mathbf{g}^\star\|_\infty$. The rate does not depend on ε, n or m, because the Lipschitz bounds do not. □

Proof of Proposition 3: After (5) with $a_i = 1$, $e^{f_i/\varepsilon} = 1/\sum_{j \le m+1} e^{(\bar{S}_{ij}+g_j)/\varepsilon}$, so $P_{i,m+1} = e^{(z+g_{m+1})/\varepsilon} \big/ \big(e^{(z+g_{m+1})/\varepsilon} + \sum_{j \le m} e^{(s_{ij}+g_j)/\varepsilon}\big)$, which equals (7). The argument of the sigmoid is strictly decreasing in each s_{ij}, because $\partial\,\mathrm{LSE}/\partial x_j > 0$, and increasing in z. □

Proof of Proposition 4: By (9), $\mathbf{V}_{j,:} = \sum_i P_{ij} \mathbf{f}_i$. Replacing P_{ij} by $c_j P_{ij}$ multiplies row j of $\mathbf{V}$ by $c_j > 0$, which intra-normalisation removes. The scene token does not depend on $\mathbf{P}$, so Φ is unchanged. For $P_{ij} = e^{(\bar{S}_{ij}+f_i+g_j)/\varepsilon}$, take $c_j = e^{-g_j/\varepsilon}$ to obtain (8), with $w_i = e^{f_i/\varepsilon}$ given by (5). For $\lambda = 0$, $\mathbf{g} = \mathbf{0}$ and $a_i = 1$. □

Proof of Proposition 5: For a similarity transform $\mathbf{u} \mapsto c\mathbf{R}\mathbf{u} + \mathbf{t}$, all pairwise distances scale by c and so does their mean μ, which leaves $\mathbf{C}^q$ unchanged. The appearance cost $\mathbf{M}$ does not depend on positions. A permutation of keypoints permutes rows and columns of $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{T}$ consistently and leaves (10) and its dust-bin constraints invariant. The mutual-arg-max correspondences, and hence their number, are therefore unchanged. □

References

[1] L. Chizat, G. Peyré, B. Schmitzer, and F.-X. Vialard, "Scaling algorithms for unbalanced optimal transport problems," *Mathematics of Computation*, vol. 87, no. 314, pp. 2563–2609, 2018.